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REGISTRATION NUMBER: 192521069

COURSE CODE: CSA1401

COURSE NAME: COMPILER DESIGN

LAB:

1.The Company ABC runs with employees with several departments. The Organization manager had all the mobile numbers of employees. Assume that you are the manager and need to verify the valid mobile numbers because there may be some invalid numbers present. Implement a LEX program to check whether the mobile number is valid or not.

CODE:

```
%{  
%}  
%%  
[6-9][0-9]{9} {printf("\nMobile Number Valid\n");}  
.+ {printf("\nMobile Number Invalid\n");}  
%%  
int yywrap(void) {}  
int main()  
{  
    printf("\nEnter Mobile Number : ");
```

```
yylex();  
printf("\n");  
}
```

OUTPUT:

```
C:\Users\pragathi>set path=C:\Users\pragathi\OneDrive\Desktop\compiler\GnuWin32\bin  
C:\Users\pragathi>flex mobile.l  
C:\Users\pragathi>set path=C:\Users\pragathi\OneDrive\Desktop\compiler\MinGW\bin  
C:\Users\pragathi>gcc lex.yy.c  
C:\Users\pragathi>a  
Enter Mobile Number : 9003152417  
Mobile Number Valid
```

2. Write a LEX Program to check the email address is valid or not.

CODE:

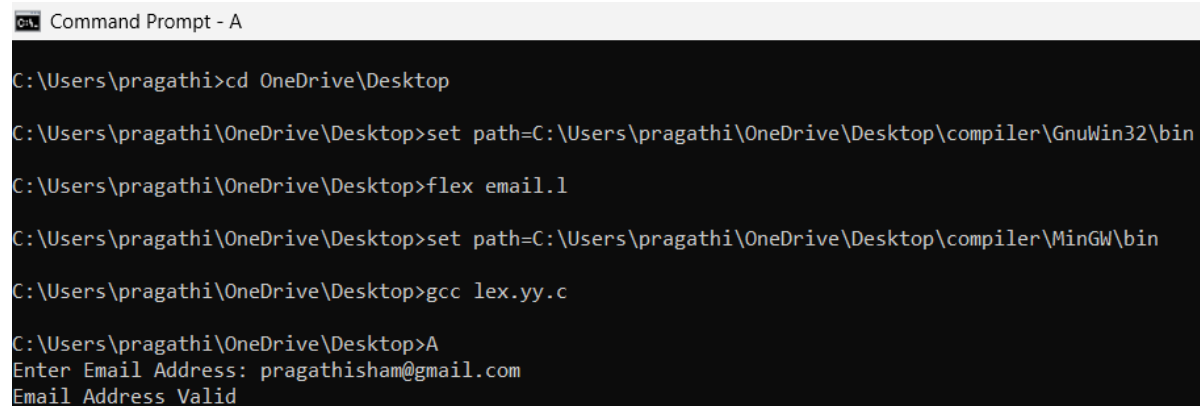
```
%{  
#include <stdio.h>  
%}  
%%  
^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$ { printf("Email Address Valid\n"); }  
.* { printf("Email Address Invalid\n"); }  
%%  
int yywrap()  
{  
    return 1;  
}
```

```

int main()
{
    printf("Enter Email Address: ");
    yylex();
    return 0;
}

```

OUTPUT:



```

C:\Users\pragathi>cd OneDrive\Desktop
C:\Users\pragathi\OneDrive\Desktop>set path=C:\Users\pragathi\OneDrive\Desktop\compiler\GnuWin32\bin
C:\Users\pragathi\OneDrive\Desktop>flex email.l
C:\Users\pragathi\OneDrive\Desktop>set path=C:\Users\pragathi\OneDrive\Desktop\compiler\MinGW\bin
C:\Users\pragathi\OneDrive\Desktop>gcc lex.yy.c
C:\Users\pragathi\OneDrive\Desktop>A
Enter Email Address: pragathisham@gmail.com
Email Address Valid

```

3. Write a LEX program to check whether the given input is a digit or not.

CODE:

```

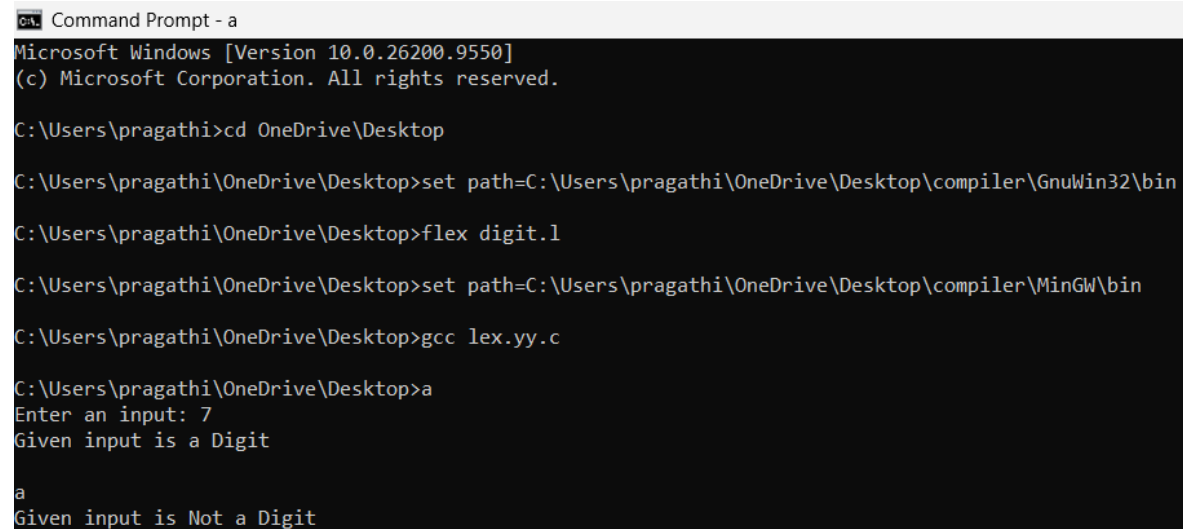
%{
#include <stdio.h>
%}
%%
^[0-9]$  { printf("Given input is a Digit\n"); }
.*      { printf("Given input is Not a Digit\n"); }
%%

```

```
int yywrap()
{
    return 1;
}

int main()
{
    printf("Enter an input: ");
    yylex();
    return 0;
}
```

OUTPUT:



```
Command Prompt - a
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pragathi>cd OneDrive\Desktop

C:\Users\pragathi\OneDrive\Desktop>set path=C:\Users\pragathi\OneDrive\Desktop\compiler\GnuWin32\bin

C:\Users\pragathi\OneDrive\Desktop>flex digit.l

C:\Users\pragathi\OneDrive\Desktop>set path=C:\Users\pragathi\OneDrive\Desktop\compiler\MinGW\bin

C:\Users\pragathi\OneDrive\Desktop>gcc lex.yy.c

C:\Users\pragathi\OneDrive\Desktop>a
Enter an input: 7
Given input is a Digit

a
Given input is Not a Digit
```